

# A Compact Narrative for Task-Visible Operator Mechanisms

Fernando Ruiz Mazo

May 1, 2026

## 1 The Question

The starting point is a simple ambiguity. A transformer trained for in-context linear regression may make predictions close to linear-kernel ridge regression, but pointwise accuracy on one sampled label vector does not identify the computation being used. Many algorithms can agree with ridge regression on the realized labels while responding differently to nearby counterfactual labels.

The note therefore changes the object of study. Instead of asking only whether the transformer predicts the right targets for one sampled context, it freezes the input geometry and perturbs only the context labels. The first question becomes:

When labels are changed infinitesimally, does the transformer respond like the ridge operator?

This operator-response test is only the entry point. The paper then asks a stronger mechanistic question: if the transformer has a ridge-like local response, is that response mediated by a compact context-side activation subspace, does the subspace have the dimension predicted by the task-visible ridge spectrum, and are the high-leverage directions in that subspace causally used?

The resulting claim is stronger than behavioral imitation but narrower than global equality. In the tested regime, the transformer locally realizes the task-distributed ridge label-response operator; that behavior is explained by a reduced KRR operator induced by activation-derived directions; the number of required directions is predicted before looking at activations; and removing the task-important directions selectively damages the response. The locality matters: the tests concern the sampled geometry, the chosen perturbation scale, and especially the task-label directions rather than all possible label perturbations.

## 2 Fixed Geometry Makes Ridge Regression an Operator

Fix one in-context episode with context and target inputs

$$X_{\text{ctx}} \in \mathbb{R}^{n_{\text{ctx}} \times d_x}, \quad X_{\text{tgt}} \in \mathbb{R}^{n_{\text{tgt}} \times d_x}.$$

The fixed geometry is

$$G = (X_{\text{ctx}}, X_{\text{tgt}}).$$

During the operator tests,  $G$  is held fixed. The context inputs do not move, the target inputs do not move, and the only varying object is the context-label vector

$$y = (y_1, \dots, y_{n_{\text{ctx}}}) \in \mathbb{R}^{n_{\text{ctx}}}.$$

Define

$$K = X_{\text{ctx}} X_{\text{ctx}}^\top, \quad K_t = X_{\text{tgt}} X_{\text{ctx}}^\top, \quad A = K + \sigma^2 I.$$

For linear-kernel ridge regression, the dual coefficients and target predictions are

$$\alpha^* = A^{-1}y, \quad y^* = K_t \alpha^* = K_t A^{-1}y.$$

Thus, once  $G$  is fixed, ridge regression is a linear map from context labels to target predictions:

$$T_G := K_t A^{-1}, \quad T_G : \mathbb{R}^{n_{\text{ctx}}} \rightarrow \mathbb{R}^{n_{\text{tgt}}}.$$

The transformer defines a map on the same spaces:

$$F_\theta^G : \mathbb{R}^{n_{\text{ctx}}} \rightarrow \mathbb{R}^{n_{\text{tgt}}}, \quad F_\theta^G(y) = \hat{y}_\theta.$$

When the geometry is clear, write  $T$  and  $F$ . The central comparison is not merely  $F(y) \approx Ty$  for one  $y$ . It is whether the local response of  $F$  to label perturbations matches the operator  $T$ .

### 3 Context-Label Directions, Local Operators, and Additivity

A context-label direction is a vector

$$u \in \mathbb{R}^{n_{\text{ctx}}}.$$

It assigns one scalar to each context example. Perturbing in direction  $u$  means changing labels as

$$y + \varepsilon u = (y_1 + \varepsilon u_1, \dots, y_{n_{\text{ctx}}} + \varepsilon u_{n_{\text{ctx}}}).$$

This is a direction over context positions. It is not a direction in input-feature space and not a direction in target space. If there are 47 context examples, these directions live in  $\mathbb{R}^{47}$ .

Pointwise agreement checks only the value of two maps at one label vector:

$$F(y) \approx Ty.$$

Since fixed-geometry ridge regression is linear, its Jacobian is the operator itself:

$$J_T(y) = T \quad \text{for every } y.$$

The transformer map  $F$  can be nonlinear in the labels. The weakest local version of “the same label-to-prediction rule” is therefore equality of first-order responses:

$$J_F(y)u \approx Tu.$$

Equivalently, for small label perturbations,

$$F(y + \delta) \approx F(y) + T\delta.$$

The experiments estimate the Jacobian action with the central finite difference

$$D_\varepsilon F(y)[u] := \frac{F(y + \varepsilon u) - F(y - \varepsilon u)}{2\varepsilon}.$$

The symmetric form is used because it cancels the leading first-order bias of a one-sided difference: for a smooth map, the error is  $O(\varepsilon^2)$  rather than  $O(\varepsilon)$ . It also treats positive and negative label perturbations in the same direction symmetrically, which is important when the goal is to estimate the local tangent response rather than a one-sided effect. The behavioral operator test asks whether

$$D_\varepsilon F(y)[u] \approx Tu$$

on held-out label directions.

This is still a local statement. Matching the value and first derivative at one point does not imply global equality. For example,  $f(x) = x + x^2$  and  $g(x) = x$  have the same value and derivative at  $x = 0$ , but differ away from zero. Thus derivative matching supports tangent-level agreement, not  $F(y') = Ty'$  for all possible labels.

The probe distribution specifies which counterfactual label directions are tested. The main probes are task-label probes:

$$u \sim \mathcal{N}(0, A).$$

This is the natural law because, under the matched linear task

$$y = X_{\text{ctx}}\beta + \eta, \quad \beta \sim \mathcal{N}(0, I), \quad \eta \sim \mathcal{N}(0, \sigma^2 I),$$

conditioning on the fixed geometry leaves only  $\beta$  and  $\eta$  random, so

$$y \mid G \sim \mathcal{N}(0, X_{\text{ctx}}X_{\text{ctx}}^\top + \sigma^2 I) = \mathcal{N}(0, A).$$

Equivalently, the conditional label covariance is

$$\text{Cov}(y \mid G) = A.$$

Thus  $u \sim \mathcal{N}(0, A)$  samples counterfactual label perturbations with the same covariance geometry as labels generated by the task, after the input geometry has been fixed. In this precise sense, task probes are the directions the task distribution itself emphasizes. In whitened coordinates, they are  $u = A^{1/2}z$ ,  $z \sim \mathcal{N}(0, I)$ . The experiments also use isotropic probes

$$u \sim \mathcal{N}(0, I),$$

which test arbitrary context-label directions equally. The distinction is central: the claim is task-local because the transformer is close to ridge on task-label probes but not on arbitrary isotropic probes. Build probes are used to construct subspaces or fitted controls; held-out probes are used for reported errors.

For a held-out probe set  $U = \{u_1, \dots, u_m\}$  and a fixed linear operator  $S : \mathbb{R}^{n_{\text{ctx}}} \rightarrow \mathbb{R}^{n_{\text{tgt}}}$ , define

$$E_\varepsilon(F, S; U) := \left( \frac{\sum_{j=1}^m \|D_\varepsilon F(y)[u_j] - Su_j\|_2^2}{\sum_{j=1}^m \|Tu_j\|_2^2 + \eta_{\text{num}}} \right)^{1/2}.$$

When  $S = T$ , this measures whether the transformer’s finite-difference response matches exact ridge regression. The denominator normalizes by the ridge-response energy on the same probes, so an error of 0.01 means a root-mean-square discrepancy of about one percent of the typical ridge response size.

**Proposition 1** (Transformer errors are not bounded by 1). *For a finite probe set  $U$ , write*

$$R_U^2 = \sum_j \|Tu_j\|_2^2.$$

*If  $D_\varepsilon F(y)[u_j] = 0$  for all probes, then*

$$E_\varepsilon(F, T; U) = \frac{R_U}{(R_U^2 + \eta_{\text{num}})^{1/2}} \leq 1.$$

If  $D_\varepsilon F(y)[u_j] = -Tu_j$ , then the same calculation gives a value at most 2. More generally, if  $D_\varepsilon F(y)[u_j] = cTu_j$ , then

$$E_\varepsilon(F, T; U) = |c - 1| \frac{R_U}{(R_U^2 + \eta_{\text{num}})^{1/2}},$$

so transformer-response errors have no universal upper bound.

*Proof.* Substitute the three proposed finite-difference responses into the definition of  $E_\varepsilon(F, T; U)$ . The last expression diverges as  $|c| \rightarrow \infty$  whenever the ridge response energy  $R_U^2$  is nonzero.  $\square$

Because ridge is linear, its higher-order derivatives are zero. We do not try to estimate all higher-order derivatives of the transformer. Instead, we check whether nonlinear interactions are small at the tested perturbation scale. Ridge has

$$T(u + v) = Tu + Tv.$$

A nonlinear transformer need not satisfy

$$D_\varepsilon F(y)[u + v] \approx D_\varepsilon F(y)[u] + D_\varepsilon F(y)[v].$$

If this approximate additivity failed badly, then  $E_\varepsilon(F, T)$  would mix two effects: failure to be ridge and failure to have a stable local linear response.

For paired directions  $V = \{(u_j, v_j)\}_{j=1}^m$ , define the second mixed finite difference

$$C_j = F(y + \varepsilon u_j + \varepsilon v_j) - F(y + \varepsilon u_j - \varepsilon v_j) - F(y - \varepsilon u_j + \varepsilon v_j) + F(y - \varepsilon u_j - \varepsilon v_j),$$

and the additivity defect

$$A_\varepsilon(F; V) := \left( \frac{\sum_{j=1}^m \|C_j\|_2^2}{\sum_{j=1}^m \|T(\varepsilon u_j + \varepsilon v_j)\|_2^2 + \eta_{\text{num}}} \right)^{1/2}.$$

For any affine function of the labels,  $C_j = 0$  exactly. Because the defect is normalized by ridge-response energy, values below one mean the non-additive interaction is smaller than the typical ridge response on the tested paired probes; values around one mean it is ridge-scale; values above one mean the interaction is larger than the ridge response, so a local-linear interpretation is unreliable. This diagnostic does not identify the operator. It only says whether a local operator comparison is meaningful at the chosen  $\varepsilon$  and probe law.

## 4 Activation Subspaces and Task-Visible Rank

The previous section gives behavioral certificates:

$$F(y) \approx Ty, \quad D_\varepsilon F(y)[u] \approx Tu.$$

These are necessary, but they are still input-output facts. They do not say whether the transformer represents the directions needed for ridge regression, or whether those directions are used. The mechanistic question is: does the residual stream expose a compact set of context-position directions sufficient for the task-visible ridge computation?

This is the right internal object to look for because ridge uses context-position dual variables. For fixed geometry,

$$\alpha^*(y) = A^{-1}y \in \mathbb{R}^{n_{\text{ctx}}}, \quad Ty = K_t \alpha^*(y).$$

The dual coordinate  $\alpha_i$  is indexed by the  $i$ -th context example. The context residual stream has the same context axis. At layer  $\ell$ ,

$$H_\ell^{\text{ctx}} \in \mathbb{R}^{n_{\text{ctx}} \times D},$$

where  $D$  is the residual width. Rows are context tokens, and the  $k$ -th column  $H_\ell^{\text{ctx}}[:, k]$  is a scalar pattern over context positions, hence a vector in  $\mathbb{R}^{n_{\text{ctx}}}$ . Linear combinations of residual-stream columns therefore define candidate directions in the same space as ridge dual variables.

Let

$$Q = [q_1, \dots, q_r] \in \mathbb{R}^{n_{\text{ctx}} \times r}$$

be a basis matrix for such candidate context-position directions. The subspace is the column span

$$\text{span}(Q) = \{Qc : c \in \mathbb{R}^r\} \subseteq \mathbb{R}^{n_{\text{ctx}}},$$

so  $Q$  is both a matrix representation and, by abuse of language, the subspace it spans. We use an  $A$ -orthonormal basis,

$$Q^\top A Q = I,$$

because the quadratic part of the ridge dual objective is  $\frac{1}{2}\alpha^\top A\alpha$ . Thus the natural size of a dual displacement is  $\alpha^\top A\alpha$ , and two directions are orthogonal in the ridge geometry when their cross-term in this quadratic energy vanishes.

Now define what it means for  $\text{span}(Q)$  to be useful for ridge. This is a restriction of the dual solution, not an ordinary restriction of the input label vector. Full ridge solves

$$\alpha^*(y) = \arg \min_{\alpha \in \mathbb{R}^{n_{\text{ctx}}}} \left\{ \frac{1}{2} \alpha^\top A \alpha - \alpha^\top y \right\},$$

whose first-order condition is  $A\alpha - y = 0$ , hence  $\alpha^* = A^{-1}y$ . To restrict the dual solution to  $\text{span}(Q)$ , write  $\alpha = Qc$ . Plugging this into the same dual objective gives

$$\frac{1}{2} c^\top Q^\top A Q c - c^\top Q^\top y.$$

Since  $Q^\top A Q = I$ , this is

$$\frac{1}{2} c^\top c - c^\top Q^\top y.$$

Differentiating with respect to  $c$  gives  $c - Q^\top y = 0$ , so

$$c^* = Q^\top y.$$

Therefore the restricted dual solution is

$$\alpha_Q(y) = Q Q^\top y.$$

Without the  $A$ -orthonormal convention, the same calculation would give

$$\alpha_Q(y) = Q(Q^\top A Q)^{-1} Q^\top y.$$

Predictions are still read out through the context-target kernel matrix  $K_t$ , so the reduced ridge operator induced by  $Q$  is

$$T_Q := K_t Q Q^\top.$$

This is the bridge from residual-stream directions to an operator-level certificate. We are not fitting an arbitrary probe. We are asking whether the directions exposed by the residual stream are enough so that ridge, restricted to those directions, still predicts like full ridge.

For two linear operators, define

$$E(S_1, S_2; U) := \left( \frac{\sum_{j=1}^m \|(S_1 - S_2)u_j\|_2^2}{\sum_{j=1}^m \|Tu_j\|_2^2 + \eta_{\text{num}}} \right)^{1/2}.$$

Thus  $E(T_Q, T; U)$  is the reduced-operator error: it measures how much prediction response is lost when full ridge is restricted to  $\text{span}(Q)$  on the tested label directions. A small value says that  $Q$  is sufficient for ridge in this operator sense. It does not yet say that the transformer uses  $Q$ .

The reason this sufficiency test is task-visible rather than global is that not every label direction matters equally for prediction. Conditional on the fixed geometry,

$$y \mid G \sim \mathcal{N}(0, A).$$

So the task naturally emphasizes perturbations

$$u \sim \mathcal{N}(0, A),$$

not arbitrary Euclidean directions. A direction is task-relevant when it is likely under this covariance geometry and when ridge produces a non-negligible target response along it. This leads to a purely ridge-theoretic benchmark: before looking inside the transformer, ask how many directions are needed by the best possible reduced ridge response on task-distributed labels.

For a rank- $r$  linear operator  $S_r : \mathbb{R}^{n_{\text{ctx}}} \rightarrow \mathbb{R}^{n_{\text{tgt}}}$ , define

$$\mathcal{E}_r(G)^2 := \inf_{\text{rank}(S_r) \leq r} \frac{\mathbb{E}_{u \sim \mathcal{N}(0, A)} \|Tu - S_ru\|_2^2}{\mathbb{E}_{u \sim \mathcal{N}(0, A)} \|Tu\|_2^2 + \eta_{\text{num}}}.$$

This asks: if only  $r$  independent label directions can affect the target prediction, what is the smallest relative mean squared prediction error one could achieve on task-label perturbations?

To compute this number, whiten the task-label perturbations:

$$u = A^{1/2}z, \quad z \sim \mathcal{N}(0, I).$$

Then

$$Tu = K_t A^{-1} A^{1/2} z = K_t A^{-1/2} z.$$

Define

$$C_G := K_t A^{-1/2}.$$

This is the ridge response in coordinates where task-label perturbations are isotropic. Singular values of  $T$  would answer the wrong question: they measure response size for raw isotropic directions  $u \sim \mathcal{N}(0, I)$ . Singular values of  $C_G$  measure response size for the task law  $u \sim \mathcal{N}(0, A)$ .

Let

$$C_G = U \text{diag}(s_i) V^\top, \quad s_1 \geq s_2 \geq \dots \geq 0.$$

In whitened direction  $v_i$ , the ridge prediction changes by  $s_i u_i$ , so  $s_i^2$  is the squared prediction-change size associated with that task direction. Since  $z$  is isotropic,

$$\mathbb{E} \|C_G z\|_2^2 = \sum_i s_i^2.$$

The best rank- $r$  approximation keeps the  $r$  largest singular directions, and the omitted prediction error is the tail:

$$\mathcal{E}_r(G)^2 = \frac{\sum_{i>r} s_i^2}{\sum_i s_i^2 + \eta_{\text{num}}}.$$

Equivalently, the numerator is the smallest possible average squared prediction error after keeping only  $r$  task-visible directions, and the denominator is the average squared ridge prediction change.

**Definition 1** (Task-visible rank). *For fixed geometry  $G$  and tolerance  $\epsilon > 0$ , define*

$$r_{\text{task}}(G; \epsilon) := \min \left\{ r : \frac{\sum_{i>r} s_i^2}{\sum_i s_i^2 + \eta_{\text{num}}} \leq \epsilon^2 \right\}.$$

Thus  $r_{\text{task}}$  is not the algebraic rank of  $T$ . It is the number of ridge-relevant directions needed so that the best reduced response has relative mean squared prediction error at most  $\epsilon^2$  on the task-label distribution. If  $\dim \text{span}(Q) < r_{\text{task}}$ , then no rank- $\dim \text{span}(Q)$  operator, even an optimally aligned one, can reach that tolerance on task probes. If  $\dim \text{span}(Q) \geq r_{\text{task}}$ , success is still not automatic: the span must also align with the task-visible ridge directions.

For the linear kernel,

$$\text{rank}(C_G) = \text{rank}(T) = \text{rank}(K_t) \leq \min\{d_x, n_{\text{ctx}}, n_{\text{tgt}}\},$$

because  $A$  is invertible and  $K_t = X_{\text{tgt}} X_{\text{ctx}}^\top$ . Hence

$$r_{\text{task}}(G; \epsilon) \leq \min\{d_x, n_{\text{ctx}}, n_{\text{tgt}}\}.$$

In the flat random linear geometries used in the target-rank sweep, this upper bound is typically attained at the reported tolerance. Writing  $X_{\text{ctx}} = P \Sigma V^\top$ ,

$$C_G = X_{\text{tgt}} V \text{diag} \left( \frac{\sigma_i}{\sqrt{\sigma_i^2 + \sigma^2}} \right) P^\top.$$

With  $n_{\text{ctx}} = 47$  and  $\sigma^2 = 0.1$ , the context singular values are typically much larger than the ridge noise, so the shrinkage factors are close to one. When  $d_x \leq n_{\text{tgt}} = 16$ , the target-side factor  $X_{\text{tgt}} V$  is generically full column rank, so the task-visible rank is empirically close to  $d_x$ . In the  $d_x = 30$  stress case,  $n_{\text{tgt}} = 16$  caps the task-visible rank.

## 5 Nonlinear Kernels and the Sample-Size Elbow

The task-visible rank definition does not rely on the linear kernel. For a general kernel, replace

$$K = X_{\text{ctx}} X_{\text{ctx}}^\top, \quad K_t = X_{\text{tgt}} X_{\text{ctx}}^\top$$

by  $K_{ij} = k(x_i, x_j)$  and  $(K_t)_{aj} = k(x_a^{\text{tgt}}, x_j)$ , keep  $A = K + \sigma^2 I$ , and again form

$$C_G = K_t A^{-1/2}.$$

The singular spectrum of  $C_G$  is the finite-episode ridge response spectrum on task-distributed label perturbations. Its tail defines the task-visible rank; write this rank as  $r_T(G; \epsilon) = r_{\text{task}}(G; \epsilon)$ . It is the number of directions a mechanism must represent, and align with, to implement KRR at tolerance  $\epsilon$  on that episode.

For nonlinear kernels,  $r_T$  should not be read as an input-dimension count. The finite matrix still obeys

$$\text{rank}(C_G) \leq \min\{n_{\text{ctx}}, n_{\text{tgt}}\},$$

but the population source of the rank is the Mercer spectrum of the kernel under the input law  $\rho$ :

$$k(x, x') = \sum_{j \geq 0} \mu_j \phi_j(x) \phi_j(x'), \quad \mu_0 \geq \mu_1 \geq \dots \geq 0.$$

With  $n = n_{\text{ctx}}$ , ridge resolves mode  $j$  with visible energy approximately

$$e_j(n) \asymp \mu_j \frac{n\mu_j}{n\mu_j + \sigma^2} = \frac{\mu_j^2}{\mu_j + \sigma^2/n}.$$

The multiplier  $n\mu_j/(n\mu_j + \sigma^2)$  is a statistical visibility factor. Modes satisfying

$$n\mu_j \ll \sigma^2$$

are below the ridge noise floor and contribute only weakly to the label-response operator. Modes satisfying  $n\mu_j \gtrsim \sigma^2$  are visible. Thus the population analogue of the task-visible rank is

$$r_*(n; \epsilon) := \min \left\{ r : \frac{\sum_{j>r} e_j(n)}{\sum_j e_j(n)} \leq \epsilon^2 \right\}.$$

This gives the right reading of finite ranks:

$$r_T(G; \epsilon) \approx \min\{n_{\text{ctx}}, n_{\text{tgt}}, r_*(n_{\text{ctx}}; \epsilon)\},$$

up to finite-sample fluctuations. The first two terms are sequence-design caps. They are not intrinsic limitations of the kernel; they can be raised by using more context or target points. The third term is intrinsic to the chosen kernel, input distribution, noise level, context size, and tolerance. It is the rank of the population-visible KRR operator.

This is the sample-size elbow. Increasing  $n_{\text{ctx}}$  lowers the visibility threshold  $\sigma^2/n_{\text{ctx}}$  and can reveal additional Mercer modes. Increasing  $n_{\text{tgt}}$  does not reveal new population modes, but it prevents the finite diagnostic from being target-capped. To train or certify a transformer as implementing the full visible KRR operator in a regime, one should choose sequence lengths so that

$$\min\{n_{\text{ctx}}, n_{\text{tgt}}, r_*(n_{\text{ctx}}; \epsilon)\} = r_*(n_{\text{ctx}}; \epsilon).$$

Otherwise the model may succeed on, and the diagnostics may certify only, a capped low-rank slice of KRR rather than the full task-visible operator. Since  $r_*(n; \epsilon)$  depends only on the kernel spectrum, input law, noise level, context size, and tolerance, it can be estimated before training and used to choose the context and target lengths.

**Proposition 2** (Task-probe scale for  $T_Q$ ). *Let  $u = A^{1/2}z$ ,  $z \sim \mathcal{N}(0, I)$ , and let  $Q^\top A Q = I$ . With*

$$C = K_t A^{-1/2}, \quad W = A^{1/2} Q, \quad P_Q = W W^\top,$$

*the population task-probe error satisfies*

$$E_{\text{task}}(T_Q, T)^2 = \frac{\|C(I - P_Q)\|_F^2}{\|C\|_F^2 + \eta_{\text{num}}}.$$



Hence

$$0 \leq E_{\text{task}}(T_Q, T) \leq 1.$$

If  $W$  spans the top  $r$  right singular directions of  $C$ , then

$$E_{\text{task}}(T_Q, T)^2 = \frac{\sum_{i>r} s_i^2}{\sum_i s_i^2 + \eta_{\text{num}}}.$$

*Proof.* Since  $W = A^{1/2}Q$ ,  $W^\top W = Q^\top A Q = I$ . For  $u = A^{1/2}z$ ,

$$Tu = K_t A^{-1} A^{1/2} z = Cz, \quad T_Q u = K_t Q Q^\top A^{1/2} z = K_t A^{-1/2} W W^\top z = C P_Q z.$$

Taking expectation over  $z$  gives

$$\mathbb{E}\|(T - T_Q)u\|_2^2 = \|C(I - P_Q)\|_F^2, \quad \mathbb{E}\|Tu\|_2^2 = \|C\|_F^2.$$

Right multiplication by the projection  $I - P_Q$  cannot increase Frobenius norm, so the error is at most 1. The singular-value expression follows from Eckart–Young: among all rank- $r$  right projections, the top right singular subspace of  $C$  minimizes the Frobenius residual, leaving  $\sum_{i>r} s_i^2$ .  $\square$

**Proposition 3** (Triangle reading). *For any fixed operator  $S$ ,*

$$E_\varepsilon(F, T; U) \leq E_\varepsilon(F, S; U) + E(S, T; U).$$

*Taking  $S = T_Q$  is the decomposition used in the experiments.*

*Proof.* Stack the probe responses as columns. The three numerators are Frobenius norms of  $M_F - M_T$ ,  $M_F - M_S$ , and  $M_S - M_T$ , and all have the same denominator  $(\sum_j \|Tu_j\|_2^2 + \eta_{\text{num}})^{1/2}$ . The result is the Frobenius triangle inequality.  $\square$

**Remark 1** (Reading the numbers). *For population task-probe reduced operators,  $E(T_Q, T) \in [0, 1]$ . An error  $e$  is a relative RMS operator error, while  $e^2$  is the missed fraction of task-probe ridge-response energy. Thus 0.01 is about one percent RMS error, 0.1 misses about one percent of squared response energy, and 0.54 misses about 29%. Values near 1 mean the reduced operator leaves almost all task-visible ridge response unexplained. Transformer errors  $E_\varepsilon(F, T)$  and  $E_\varepsilon(F, T_Q)$  are not bounded by 1, as noted above.*

The mechanism argument therefore needs three distinct checks:

$$E_\varepsilon(F, T; U) \text{ small}, \quad E(T_Q, T; U) \text{ small}, \quad E_\varepsilon(F, T_Q; U) \text{ small}.$$

The first is behavioral. The second says the residual-stream subspace supports ridge in the reduced-operator sense. The third says the transformer’s local response follows the reduced operator induced by that same subspace. The decomposition

$$D_\varepsilon F(y)[u] - Tu = (D_\varepsilon F(y)[u] - T_Q u) + (T_Q u - Tu)$$

keeps these claims separate.

It remains to describe how the residual-stream subspace is extracted. The experiments use two final-state constructions in this section, and the next section introduces the causal layerwise construction. Given any context-pattern matrix  $M \in \mathbb{R}^{n_{\text{ctx}} \times p}$ , compute the  $A$ -weighted SVD

$$A^{1/2}M = U\Sigma V^\top,$$

keep the selected left singular directions, and map back:

$$Q(M) := A^{-1/2}U_{\text{kept}}.$$

Then  $Q(M)^\top A Q(M) = I$ . The kept singular directions are those above the numerical threshold or within the predeclared rank budget. The task-visible rank tells how many ridge-relevant directions should be enough in principle; the experiments then ask whether the residual-stream directions achieve small  $E(T_Q, T)$  around that scale.

The raw final-state extractor uses the unperturbed final residual stream,

$$M_{\text{raw}} = H_L^{\text{ctx}}, \quad Q_{\text{raw}} := Q(M_{\text{raw}}).$$

It asks an availability question: which context-position directions are linearly visible in the final residual stream, and are enough of them present to support the task-visible ridge operator?

The response-only extractor uses finite-difference changes in the final residual stream,

$$Z_L(u) = \frac{H_L^{\text{ctx}}(y + \varepsilon u) - H_L^{\text{ctx}}(y - \varepsilon u)}{2\varepsilon}, \quad M_{\text{resp}} = [Z_L(u_1), \dots, Z_L(u_m)],$$

and sets

$$Q_{\text{resp}} := Q(M_{\text{resp}}).$$

It asks the stricter question: among the directions that actually move when labels are perturbed, is there a context-position subspace whose induced  $T_Q$  is close to ridge?

In both constructions  $Q$  is frozen after extraction. If the basis were re-extracted for each held-out perturbation, it would not define one stable reduced operator. The certificate is that a fixed subspace extracted from residual-stream activations induces a fixed  $T_Q$  that agrees with ridge and with the transformer’s held-out local response.

## 6 Causal Layerwise Construction

The final-state certificate can show that a useful subspace is present at the end of the computation, but it does not explain how that subspace appears across layers. The layerwise construction is a post-training diagnostic for this question. It looks for context-side directions that satisfy three conditions: they move when labels are perturbed, they are new relative to what earlier layers have already supplied, and they are causally useful for the transformer’s response.

The reported dimension counts only directions that are directly accepted from residual-stream responses.

At layer  $\ell$ , define the hidden-state finite-difference response

$$Z_\ell(u) := \frac{H_\ell^{\text{ctx}}(y + \varepsilon u) - H_\ell^{\text{ctx}}(y - \varepsilon u)}{2\varepsilon}.$$

Stacking these matrices over build probes gives

$$M_\ell = [Z_\ell(u_1), \dots, Z_\ell(u_m)].$$

The columns of  $M_\ell$  are context-position patterns that are visible at layer  $\ell$  when context labels move. This is the first filter: a direction that does not appear in label-induced hidden-state changes is not counted as part of the label-computation mechanism.

The second filter prevents double-counting. Define the accepted direction matrices inductively. Initialize with no accepted directions. At the beginning of layer  $\ell$ , assume that

$$S_0, \dots, S_{\ell-1}$$

have already been defined, where  $S_j$  is the matrix of directions first accepted at layer  $j$  and is empty if none were accepted. Declare as already available the seed span generated by these earlier accepted directions:

$$R_\ell^{\text{pre}} := \text{span}\{S_j : 0 \leq j < \ell\},$$

where the span is over the columns of these matrices. For  $\ell = 0$ , this span is empty. Projection and orthogonality are measured in the  $A$ -geometry, because the candidate directions are later used to define the reduced ridge operator.

Let  $Q_\ell^{\text{pre}}$  be an  $A$ -orthonormal basis for this pre-existing span. The novel component of the layer response is

$$N_\ell = \left( I - Q_\ell^{\text{pre}} (Q_\ell^{\text{pre}})^\top A \right) M_\ell.$$

Thus  $N_\ell$  is the label-sensitive layer response after removing directions already accepted from earlier layers. Later layers do not receive credit for re-exposing the same context-position direction.

An  $A$ -weighted SVD of  $N_\ell$  proposes candidate directions

$$A^{1/2} N_\ell = U_\ell \Sigma_\ell V_\ell^\top, \quad q_i = A^{-1/2} U_{\ell,i}.$$

These candidates are visible and novel, but visibility is still not use.

The third filter is causal. For each candidate  $q_i$ , remove its  $A$ -orthogonal projection from the context residual stream at layer  $\ell$ :

$$H_\ell^{\text{ctx}} \mapsto H_\ell^{\text{ctx}} - q_i q_i^\top A H_\ell^{\text{ctx}},$$

roll the remaining layers forward, and measure the damage to the finite-difference response on a separate causal-probe set. A candidate is accepted only if this damage exceeds a predeclared threshold. Let  $I_\ell$  be the accepted candidate indices. This completes the induction step by defining

$$S_\ell := A^{-1/2} U_{\ell, I_\ell}.$$

If  $I_\ell$  is empty, then  $S_\ell$  is empty. Otherwise its columns are  $A$ -orthonormal and are the directions first accepted at layer  $\ell$ . The selection rule is deliberately stronger than decodability: a direction must be visible, novel, and functionally consequential at the tested layer.

After processing all layers, the accepted directions define the seed layerwise activation span

$$R_{\text{seed}} := \text{span}\{S_\ell : 0 \leq \ell \leq L\}, \quad d_{\text{seed}} := \dim R_{\text{seed}}.$$

Let  $Q_{\text{seed}}$  be an  $A$ -orthonormal basis of this span. The induced reduced operator is

$$T_{Q_{\text{seed}}} = K_t Q_{\text{seed}} Q_{\text{seed}}^\top.$$

The layerwise test then compares  $d_{\text{seed}}$  to the task-visible rank and compares  $T_{Q_{\text{seed}}}$  to  $T$ . The dimension  $d_{\text{seed}}$  is an accounting statistic, not the certificate by itself. It says how many directly accepted context-position directions the checkpoint exposes. The operator error  $E(T_{Q_{\text{seed}}}, T)$  says whether those directions are aligned with the task-visible ridge directions. A span may have enough dimension and still miss part of  $T$ , and a full-dimensional span can make  $T_Q = T$  trivially without implying that the transformer behaves like ridge.

## 7 Experiment 1: Final-State Operator Certificate

Experiment 1 is the endpoint certificate. It asks two questions. First, does the trained transformer locally behave like exact KRR at the final prediction output? Second, if it does, is there a compact final-state context-position subspace whose induced reduced ridge operator explains that behavior?

For each episode, the geometry  $G$  is fixed and the exact ridge operator

$$T = K_t A^{-1}$$

is computed. The experiment draws separate build probes and held-out evaluation probes. Build probes are used only to extract bases or fit controls; all reported operator errors are evaluated after the basis is frozen. In the reported run there are 4 episodes, 16 build probes, 32 held-out evaluation probes, finite-difference scale  $\varepsilon = 10^{-3}$ , and a predeclared extraction rank cap  $r_{\max} = 9$ .

The experiment constructs two final-state bases. The raw basis uses the unperturbed final context residual stream

$$M_{\text{raw}} := H_L^{\text{ctx}} \in \mathbb{R}^{n_{\text{ctx}} \times D}.$$

Each residual coordinate is a context-position pattern, and the  $A$ -weighted SVD of  $M_{\text{raw}}$  gives an  $A$ -orthonormal basis  $Q_{\text{raw}}$ .

The response-only basis uses only the part of the final residual stream that moves under label perturbations. For task-label build probes  $u_1, \dots, u_m$ , define

$$Z_L(u_j) = \frac{H_L^{\text{ctx}}(y + \varepsilon u_j) - H_L^{\text{ctx}}(y - \varepsilon u_j)}{2\varepsilon}, \quad M_{\text{resp}} = [Z_L(u_1), \dots, Z_L(u_m)].$$

An  $A$ -weighted SVD of  $M_{\text{resp}}$  gives  $Q_{\text{resp}}$ . This is a stricter availability test than the raw basis: the directions must be visible in label-induced hidden-state changes, not merely present in the final residual stream.

For either frozen basis  $Q$ , the KRR-structured reduced operator is

$$T_Q = K_t Q Q^\top.$$

The experiment evaluates both subspace sufficiency,

$$E(T_Q, T),$$

and model-subspace alignment,

$$E_\varepsilon(F, T_Q).$$

It also includes a same-span least-squares readout control, denoted  $T_{\text{actLR}}$ . For any perturbation  $u$ , both  $T_Q$  and this control first form the same activation coordinates

$$z = Q^\top u.$$

The KRR-structured reduced operator then uses the KRR-implied readout from those coordinates:

$$T_Q u = (K_t Q)(Q^\top u), \quad C_{\text{KRR}} = K_t Q.$$

The control keeps the same features  $Q^\top u$ , but fits the readout directly to the transformer's finite-difference responses on build probes  $u_1, \dots, u_B$ :

$$C_Q \in \arg \min_{C \in \mathbb{R}^{n_{\text{tgt}} \times r}} \sum_{j=1}^B \left\| D_\varepsilon F(y)[u_j] - C Q^\top u_j \right\|_2^2.$$

It then defines

$$T_{\text{actLR}}u = C_Q(Q^\top u), \quad T_{\text{actLR}} := C_Q Q^\top.$$

Thus the two operators use the same  $Q$ -features but different readouts:  $T_Q$  uses the fixed KRR readout  $K_t Q$ , while  $T_{\text{actLR}}$  uses the least-squares readout  $C_Q$  fitted to the transformer. This control is important because it separates two questions. A good  $T_{\text{actLR}}$  fit would show that the transformer’s local response is decodable from the activation coordinates. A good  $T_Q$  fit shows something stronger: the same coordinates support the KRR-constrained readout.

The behavioral check succeeds on task-label probes. The pointwise prediction is already close to KRR, with relative model-KRR discrepancy 0.01045, and the held-out finite-difference operator error is

$$E_\varepsilon(F, T) = 0.0121.$$

The additivity defect is much smaller than this error:  $4.43 \times 10^{-4}$  on task probes and  $3.49 \times 10^{-3}$  on isotropic probes at  $\varepsilon = 10^{-3}$ , with comparable values at  $10\varepsilon$ . Thus the local response is sufficiently close to additive for the operator comparison to be meaningful.

The final-state subspace certificates also succeed on task probes. For the raw basis,

$$E(T_Q, T) = 0.00148, \quad E_\varepsilon(F, T_Q) = 0.01194.$$

For the response-only basis,

$$E(T_Q, T) = 0.00367, \quad E_\varepsilon(F, T_Q) = 0.01239.$$

So the final residual stream exposes compact context-position subspaces whose induced reduced ridge operators are close to exact KRR, and the transformer’s local response follows those reduced operators.

The  $T_{\text{actLR}}$  control shows what this certificate is not saying. On task probes,  $T_{\text{actLR}}$  fits the transformer slightly better than  $T_Q$ : for the raw basis

$$E_\varepsilon(F, T_{\text{actLR}}) = 0.00726 \quad \text{versus} \quad E_\varepsilon(F, T_Q) = 0.01194.$$

But it agrees worse with exact KRR:

$$E(T_{\text{actLR}}, T) = 0.01202 \quad \text{versus} \quad E(T_Q, T) = 0.00148.$$

The response-only basis shows the same pattern. Therefore the certificate is not merely that a free low-rank readout can fit the model response; it is that the same activation-derived subspace supports the KRR-structured operator. Matched random subspaces of the same rank fail the operator-sufficiency test by large factors on task probes, with  $E(T_Q, T) \approx 0.117$  for the raw rank and 0.0955 for the response rank.

The limitation appears immediately off distribution. On isotropic probes,

$$E_\varepsilon(F, T) \approx 0.302.$$

The raw reduced operator remains closer to exact KRR than the transformer response does,

$$E(T_Q, T) = 0.0525, \quad E_\varepsilon(F, T_Q) = 0.2915,$$

but the model is no longer KRR-like on arbitrary label directions. Experiment 1 therefore supports a task-local final-state operator certificate, not a global implementation of ridge regression on all context-label directions.

Table 1: Experiment 1 predictive and behavioral checks, mean over 4 episodes.

| Quantity                                           | Transformer | Exact KRR | Transformer-KRR |
|----------------------------------------------------|-------------|-----------|-----------------|
| Pointwise MSE to noiseless teacher targets         | 0.0161      | 0.0148    | –               |
| Pointwise relative discrepancy                     | –           | –         | 0.01045         |
| Held-out task operator error $E_\varepsilon(F, T)$ | –           | –         | 0.0121          |

Table 2: Experiment 1 additivity defect, mean over 4 episodes. The defect measures the normalized failure of finite-difference responses to add:  $(\sum_j \|D_\varepsilon F[u_j + v_j] - D_\varepsilon F[u_j] - D_\varepsilon F[v_j]\|^2 / (\sum_j \|T(u_j + v_j)\|^2 + \eta_{\text{num}}))^{1/2}$ .

| Probe distribution | $\varepsilon = 10^{-3}$ | $\varepsilon = 10^{-2}$ |
|--------------------|-------------------------|-------------------------|
| Task-label probes  | $4.43 \times 10^{-4}$   | $3.52 \times 10^{-4}$   |
| Isotropic probes   | $3.49 \times 10^{-3}$   | $2.93 \times 10^{-3}$   |

Table 3: Reduced operator vs. same-span response-fitted low-rank control, mean over 4 episodes.  $E_\varepsilon(F, S)$  denotes transformer-to- $S$  finite-difference operator error;  $E(S, T)$  denotes  $S$ -to-ridge operator error.

| Probe distribution | Basis    | $E(F, T_Q)$ | $E(F, T_{\text{actLR}})$ | $E(T_Q, T)$ | $E(T_{\text{actLR}}, T)$ |
|--------------------|----------|-------------|--------------------------|-------------|--------------------------|
| Task probes        | raw      | 0.01194     | 0.00726                  | 0.00148     | 0.01202                  |
| Task probes        | response | 0.01239     | 0.01169                  | 0.00367     | 0.01645                  |
| Isotropic probes   | raw      | 0.2915      | 0.2007                   | 0.05254     | 0.2687                   |
| Isotropic probes   | response | 0.3193      | 0.2418                   | 0.1278      | 0.2929                   |

The three plots below should be read as diagnostic decompositions of the same certificate. In all of them the vertical axis is a relative error on a logarithmic scale, so lower is better and equal vertical gaps correspond to multiplicative changes. The word “label” in the plot labels means task-label probes,  $u \sim N(0, A)$ ; “isotropic” means  $u \sim N(0, I)$ . The first plot asks whether a free response fit from the same coordinates gives the same conclusion as the KRR-constrained reduced operator. The second plot asks how the certificate changes as we move continuously from task-label probes to isotropic probes. The third plot asks how much retained rank is needed before the final-state basis supports the task-visible ridge computation.

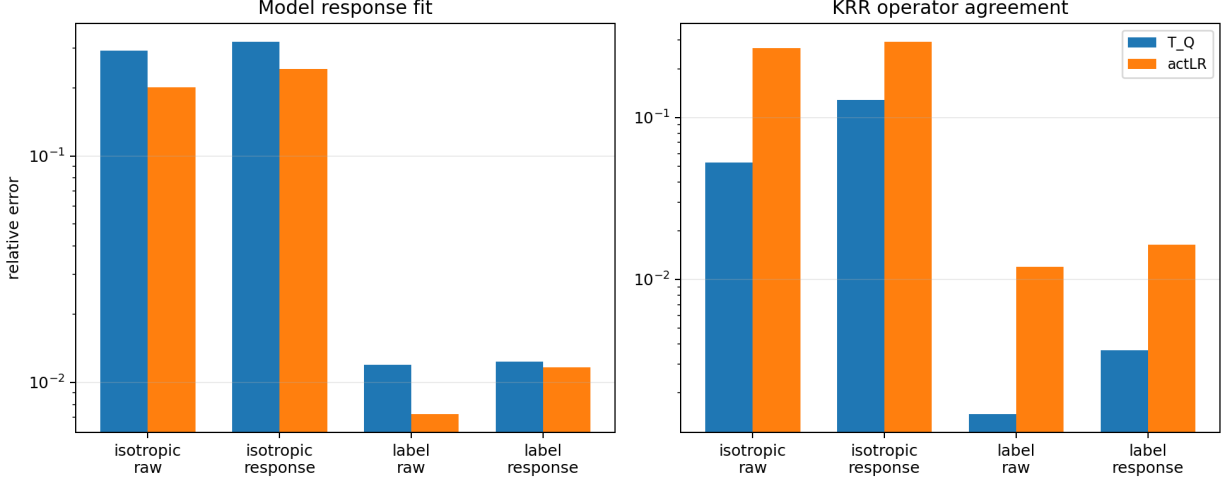


Figure 1: Experiment 1 reduced-operator certificate versus same-span response fitting. Left panel: transformer-to-operator response error  $E_\varepsilon(F, S)$ , where  $S = T_Q$  for the KRR-constrained reduced operator and  $S = T_{actLR}$  for the same-span response-fitted control. This panel measures only how well the candidate operator predicts the transformer’s finite-difference responses. Right panel: operator-to-ridge error  $E(S, T)$ , which measures whether the candidate operator is actually close to exact KRR. Thus a good mechanistic certificate should be low in both panels. The free response-fitted operator improves the left panel on task probes, especially for the raw basis, but it is worse in the right panel; it fits the model slightly better by giving up KRR structure. The KRR-constrained  $T_Q$  is therefore the stronger ridge certificate.

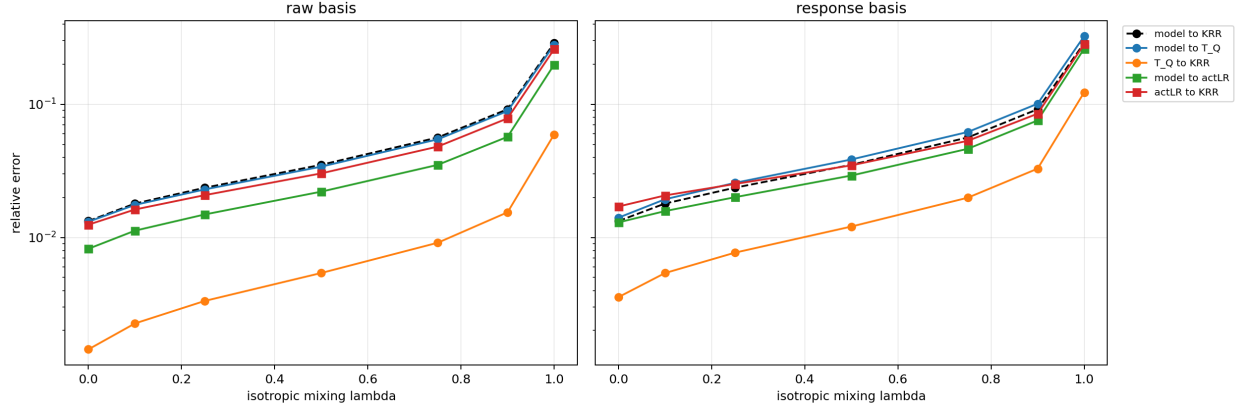


Figure 2: Experiment 1 probe interpolation. The horizontal axis is the probe mixing parameter  $\lambda$ , where probes are drawn as  $u_\lambda = \sqrt{1-\lambda}u_{\text{task}} + \sqrt{\lambda}u_{\text{iso}}$ , with  $u_{\text{task}} \sim N(0, A)$  and  $u_{\text{iso}} \sim N(0, I)$ . Thus  $\lambda = 0$  is the task-label regime and  $\lambda = 1$  is the fully isotropic regime. The black dashed curve is the direct behavioral error  $E_\varepsilon(F, T)$ . The blue and orange curves decompose the  $T_Q$  certificate into model-to- $T_Q$  error  $E_\varepsilon(F, T_Q)$  and  $T_Q$ -to-KRR error  $E(T_Q, T)$ . The green and red curves show the same two errors for the response-fitted control  $T_{\text{actLR}}$ . The key pattern is that all task-probe errors are small near  $\lambda = 0$ , but the model-response errors rise as the probes become isotropic. This is the plot-level evidence that the certificate is task-local rather than a global statement over arbitrary label directions.

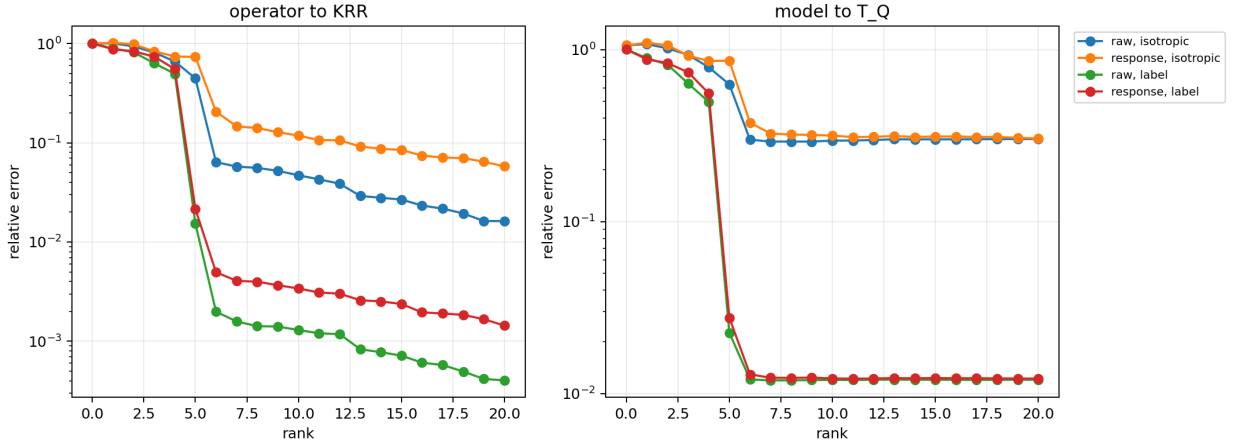


Figure 3: Experiment 1 rank curves for raw and response-only final-state bases. The horizontal axis is the retained basis rank  $r$ . At each  $r$ , the experiment keeps the first  $r$  extracted directions, forms  $T_Q = K_t Q Q^\top$ , and reevaluates the operator errors. Left panel: reduced-operator error  $E(T_Q, T)$ , so this panel asks whether the retained final-state directions are sufficient to reproduce ridge on the probe distribution. Right panel: transformer-to-reduced-operator response error  $E_\varepsilon(F, T_Q)$ , so this panel asks whether the transformer’s local response follows that reduced operator. The task-label curves drop sharply once enough directions are retained, while the isotropic curves remain substantially larger. This shows both that the final residual stream exposes the task-relevant ridge directions and that these directions do not explain arbitrary label perturbations equally well.



## 8 Experiment 2: Layerwise Span and Task-Visible Rank

The second experiment asks how the needed directions appear across layers. More precisely, it asks whether the trained transformer exposes a compact seed span of directly accepted residual-stream directions, whether that span is on the task-visible scale predicted by the KRR spectrum, and whether the reduced operator induced by that span approximates  $T$ .

At layer  $\ell$ , define the hidden finite-difference response

$$Z_\ell(u) = \frac{H_\ell^{\text{ctx}}(y + \varepsilon u) - H_\ell^{\text{ctx}}(y - \varepsilon u)}{2\varepsilon}.$$

Stacking these responses over build probes gives  $M_\ell$ , a matrix of context-position patterns visible in layer  $\ell$  when labels are perturbed.

The construction processes layers sequentially. Before layer  $\ell$ , directions already accepted from earlier layers define a pre-existing  $A$ -orthonormal span  $Q_\ell^{\text{pre}}$ . The novel part of the layer response is

$$N_\ell = \left( I - Q_\ell^{\text{pre}}(Q_\ell^{\text{pre}})^\top A \right) M_\ell.$$

An  $A$ -weighted SVD of  $N_\ell$  gives candidate directions. Visibility is not enough: a candidate is accepted only if removing it from the context residual stream at that layer changes the transformer’s finite-difference response by a predeclared amount.

The accepted directions across layers define the seed layerwise activation span

$$R_{\text{seed}} = \text{span}\{S_0, \dots, S_L\}, \quad d_{\text{seed}} = \dim R_{\text{seed}}.$$

Let  $Q_{\text{seed}}$  be an  $A$ -orthonormal basis of this span and

$$T_{Q_{\text{seed}}} = K_t Q_{\text{seed}} Q_{\text{seed}}^\top.$$

The diagnostic expectation is a post-training resource law with an alignment condition:

$$d_{\text{seed}}/r_{\text{task}} \approx 1 \quad \implies \quad E(T_{Q_{\text{seed}}}, T) \approx 0,$$

provided the selected span is aligned with the task-visible ridge directions. This is not an exact rank identity. The seed span counts directions used by the trained transformer;  $r_{\text{task}}$  is the optimal ridge rank for task-distributed perturbations. Dimension by itself is not the certificate; the reduced operator  $T_{Q_{\text{seed}}}$  is the certificate.

The target-rank sweep supports this interpretation. Here  $r_T$  denotes the tail-based task-visible rank at RMS tolerance  $\epsilon = 0.01$ , matching the definition above. For

$$d_x \in \{3, 5, 8, 10\},$$

the seed span contains slightly more directions than the empirical task-visible rank and the reduced-operator error is about  $10^{-3}$ . This is the desired pattern: the model exposes a small number of directly accepted residual-stream directions, those directions are on the same scale as the ridge rank requirement, and restricting ridge to their span loses essentially none of the task-local operator response. For  $d_x = 15$ , the seed span has enough dimension on average,

$$d_{\text{seed}} \approx 15.8, \quad r_{\text{task}} \approx 14.9,$$

but the reduced-operator error rises to

$$E(T_{Q_{\text{seed}}}, T) = 0.08493.$$

Table 4: Experiment 2 complete target-rank sweep with causal seed-span selection, mean over 16 episodes per setting.

| Setting    | $d_{\text{seed}}$ | $r_T$ | $d_{\text{seed}}/r_T$ | $E(T_Q, T)$ | $E(F, T_Q)$ | $E(F, T)$ |
|------------|-------------------|-------|-----------------------|-------------|-------------|-----------|
| $d_x = 3$  | 4.6               | 3.0   | 1.542                 | 0.00114     | 0.00641     | 0.00629   |
| $d_x = 5$  | 6.0               | 5.0   | 1.200                 | 0.00135     | 0.00926     | 0.00911   |
| $d_x = 8$  | 9.0               | 8.0   | 1.125                 | 0.00174     | 0.01317     | 0.01293   |
| $d_x = 10$ | 11.0              | 10.0  | 1.100                 | 0.00135     | 0.01309     | 0.01305   |
| $d_x = 15$ | 15.8              | 14.9  | 1.063                 | 0.08493     | 0.09598     | 0.01743   |
| $d_x = 30$ | 47.0              | 16.0  | 2.938                 | 0.00000     | 1.68673     | 1.68673   |

The failure is therefore not just a rank shortage. It says that the accepted seed directions have enough dimension on average, but do not consistently span all task-visible ridge directions.

The  $d_x = 30$  row is an exploratory stress case. Since  $n_{\text{tgt}} = 16$ , the task-visible rank is capped at  $r_T = 16$ , even though the input dimension is 30. In the seed-span rerun, causal selection expands to the full context-position space,  $d_{\text{seed}} = 47$ , so the reduced operator equals exact KRR by construction:

$$E(T_Q, T) = 0.$$

But the model itself is not behaviorally close to ridge:

$$E(F, T) = 1.68673.$$

This is a useful stress-test warning. A large or even full seed span can make  $T_Q$  ridge-sufficient without showing that the transformer uses that operator. The model-subspace alignment error

$$E(F, T_Q) = 1.68673$$

exposes the failure. Thus  $d_x = 30$  reinforces the diagnostic separation: representation sufficiency is not behavioral success, and dimension alone is not the mechanism claim.

Table 5: Experiment 2 fixed- $d_x = 5$  depth/head sweep with causal seed-span selection, mean over 16 episodes. All checkpoints use width  $D = 128$ .

| Checkpoint | $L$ | $H$ | $d_{\text{seed}}$ | $r_T$ | $E(T_Q, T)$ | $E(F, T_Q)$ | $E(F, T)$ |
|------------|-----|-----|-------------------|-------|-------------|-------------|-----------|
| $L = 2$    | 2   | 4   | 16.3              | 5.0   | 0.00095     | 0.18537     | 0.18535   |
| $L = 4$    | 4   | 4   | 6.2               | 5.0   | 0.00127     | 0.02534     | 0.02528   |
| $L = 6$    | 6   | 4   | 6.1               | 5.0   | 0.00137     | 0.01272     | 0.01255   |
| $L = 8$    | 8   | 4   | 6.0               | 5.0   | 0.00147     | 0.00962     | 0.00952   |
| $L = 12$   | 12  | 4   | 6.1               | 5.0   | 0.00110     | 0.00953     | 0.00943   |
| $H = 1$    | 8   | 1   | 6.2               | 5.0   | 0.00085     | 0.01558     | 0.01559   |
| $H = 2$    | 8   | 2   | 6.1               | 5.0   | 0.00108     | 0.00908     | 0.00904   |
| $H = 8$    | 8   | 8   | 6.0               | 5.0   | 0.00164     | 0.00996     | 0.00982   |

The fixed- $d_x = 5$  depth and head sweeps add an important qualification. These checkpoints keep the task dimension and width fixed while varying depth over  $L \in \{2, 4, 6, 8, 12\}$  and head count over  $H \in \{1, 2, 4, 8\}$ . Causal seed selection gives  $E(T_Q, T) \approx 10^{-3}$  for all reported checkpoints, but behavior still varies: the shallow  $L = 2$  model has a strong reduced-operator certificate but  $E(F, T) = 0.185$ , while the  $L = 8$ ,  $H = 2$ , and  $L = 12$  checkpoints have errors near  $10^{-2}$ . Thus the sweep is a control against conflating availability with use:

$$E(T_Q, T) \text{ small}$$

does not imply

$$E_\varepsilon(F, T) \text{ small.}$$

The activations may expose a ridge-sufficient subspace even when the trained transformer does not use it accurately enough to behave like KRR.

## 9 Dimension-40 Regime-Mismatch Stress Test

The  $d_x = 40$  checkpoint is useful because it separates two questions that are easy to conflate: whether KRR-relevant directions are present in the residual stream, and whether the trained model uses them as a full target-rich KRR operator. The checkpoint was trained in a target-poor regime:

$$n_{\text{tgt}} = 8, \quad \tau(\Sigma) = \sum_{i=1}^{40} \frac{\lambda_i}{\lambda_i + \sigma^2} \leq 16,$$

with  $n_{\text{ctx}}$  sampled as a multiple of the effective dimension. Thus each training episode directly supervised only a small number of target queries. Across many episodes this is still a Monte Carlo signal for the population prediction loss, but it does not force one forward pass to expose a high-rank simultaneous target operator.

The diagnostic deliberately uses a richer geometry:

$$n_{\text{ctx}} = 47, \quad n_{\text{tgt}} = 40.$$

In this setting the task-visible rank is

$$r_T = 32.4,$$

well above the training target count. Hence the diagnostic is a stress test of whether a target-poor-trained checkpoint generalizes to a much richer finite-episode operator.

Table 6:  $d_x = 40$  target-rich diagnostics. The first two rows come from the final-state certificate run. The causal seed rows come from a separate 16-episode causal run; both arithmetic means and medians are shown because two catastrophic episodes dominate the mean. Within a single run,  $E(F, T)$  is independent of the chosen basis.

| Run         | Basis    | Statistic | $\dim Q$ | $r_T$ | $E(T_Q, T)$ | $E(F, T_Q)$ | $E(F, T)$ |
|-------------|----------|-----------|----------|-------|-------------|-------------|-----------|
| Final-state | raw      | mean      | 40.0     | –     | 0.01922     | 0.08512     | 0.08541   |
| Final-state | response | mean      | 40.0     | –     | 0.07400     | 0.11286     | 0.08541   |
| Causal seed | seed     | mean      | 29.2     | 32.4  | 0.13741     | 98.74530    | 98.67662  |
| Causal seed | seed     | median    | 28.0     | 35.0  | 0.17142     | 0.21569     | 0.12781   |

The final-state certificate shows availability. At rank 40, the raw final-state basis induces a reduced KRR operator close to exact KRR on task-label probes:

$$E(T_{Q_{\text{raw}}}, T) = 0.01922.$$

The response-only final basis is weaker but still informative,

$$E(T_{Q_{\text{resp}}}, T) = 0.07400.$$

These numbers say that the final residual stream contains many context-position directions from which a KRR-structured operator can be built. The common value of  $E(F, T)$  across the two final-state rows is expected, because  $E(F, T)$  does not depend on whether the comparison basis is raw or response-only. These final-state rows do not say that the transformer uses those directions causally, and they also use a broad rank cap 40, not the task-visible rank itself.

The causal target-rich diagnostic gives the harder reading, but the arithmetic mean must be read carefully. It accepts

$$d_{\text{seed}} = 29.2$$

directions on average, while KRR requires

$$r_T = 32.4$$

task-visible directions at the reported tolerance. The dimension ratio is therefore about 0.90. The median episode is similar in scale, with  $d_{\text{seed}} = 28.0$  and  $r_T = 35.0$ . Moreover,

$$E(T_{Q_{\text{seed}}}, T) = 0.13741,$$

with median 0.17142, so the accepted span is not merely a little short; it is also not perfectly aligned with the task-visible KRR directions.

The model-response mean is much less representative. The causal run contains two catastrophic pointwise failures: episodes 3 and 15 have model/KRR MSE ratios 3304.6 and 35537.9, respectively. These two episodes drive

$$E(F, T) = 98.67662.$$

The median is instead

$$E(F, T) = 0.12781,$$

and after excluding episodes with  $E(F, T) > 1$ , the mean over the remaining 14 episodes is

$$E(F, T) = 0.12346.$$

Thus the target-rich causal diagnostic shows lack of robustness rather than uniform failure on every episode: most episodes are moderately KRR-like, but rare scale/spectrum cases cause catastrophic breakdown.

The interpretation is not that target-poor training is theoretically incapable of learning KRR. Resampled small target sets can train the population prediction rule in expectation. The point is mechanistic and operational: training with  $n_{\text{tgt}} = 8$  did not give strong per-episode pressure to build a reusable  $r_T \approx 32$  label-response operator. When the diagnostic removes the target cap, it asks for a richer simultaneous computation than the checkpoint was directly trained to expose. The  $d_x = 40$  result should therefore be read as a regime-mismatch stress test: the model contains substantial KRR-sufficient final-state structure, but the causal seed span is short or misaligned relative to the target-rich  $r_T$ , and the model has rare catastrophic failures in this richer regime.

This motivates the sequence-length prescription above. If the goal is to train and certify a true task-visible KRR mechanism in a given regime, choose  $n_{\text{ctx}}$  and  $n_{\text{tgt}}$  so that

$$\min\{n_{\text{ctx}}, n_{\text{tgt}}, r_*(n_{\text{ctx}}; \epsilon)\} = r_*(n_{\text{ctx}}; \epsilon)$$

during training as well as diagnosis. Target-rich diagnostics alone can reveal the missing operator; target-rich training is the direct way to put pressure on the model to learn it.

## 10 RBF Kernel Experiment: Prediction-Risk Rank and Availability

The RBF experiment tests whether the same operator picture survives when the task-visible rank is no longer tied to the input dimension. The checkpoint is trained on labels from an RBF Gaussian process with lengthscale 3.0, signal variance 1.0, and noise variance  $\sigma^2 = 0.1$ . On held-out RBF tasks the model is close to KRR: at the end of training the prediction MSE is within about 1–3% of exact KRR, and the model–KRR prediction discrepancy is around  $10^{-3}$ .

For this nonlinear task, the strict rank used above,

$$r_{1\%op} = \min \left\{ r : \sqrt{\frac{\sum_{i>r} s_i^2}{\sum_i s_i^2 + \eta_{\text{num}}}} \leq 0.01 \right\},$$

is a deliberately stringent operator-completion diagnostic. It asks for one-percent relative operator recovery even when the omitted modes are below the task’s prediction noise scale. For the RBF task we therefore also compute the prediction-risk rank

$$r_{5\%} = \min \left\{ r : \sum_{i>r} s_i^2 \leq 0.05 R_{\text{KRR}} \right\}, \quad R_{\text{KRR}} = \text{tr} \left( K_{tt} - K_t A^{-1} K_t^\top \right).$$

This is model-independent: it is computed from the KRR geometry before looking at activations. It asks how many modes are needed so that discarding the rest would add at most five percent of the KRR posterior prediction risk. Equivalently, it induces an episode-dependent operator tolerance

$$\tau_{\text{task}} = \sqrt{\frac{0.05 R_{\text{KRR}}}{\sum_i s_i^2 + \eta_{\text{num}}}}.$$

In the RBF spectral scan this tolerance is much looser than 0.01: about 0.064 for  $n_{\text{ctx}} = 47$  and about 0.044 for  $n_{\text{ctx}} = 128$ . This is the missing normalization. The strict rank counts low-energy tail modes;  $r_{5\%}$  counts modes that matter relative to the irreducible KRR prediction risk.

Table 7: RBF diagnostics.  $r_{5\%}$  is the prediction-risk rank;  $r_{1\%op}$  is the old strict relative-operator rank. The final-state column reports the raw final-basis reduced-operator error at the nearest rank to  $r_{5\%}$ . The causal seed columns use the layerwise causal extractor and should be read as a stricter mechanistic test, not as the definition of task rank.

| Regime                                       | $r_{5\%}$ | $r_{1\%op}$ | raw final | $E(T_Q, T)$ | $d_{\text{seed}}$ | seed | $E(T_Q, T)$ | MSE/KRR |
|----------------------------------------------|-----------|-------------|-----------|-------------|-------------------|------|-------------|---------|
| $n_{\text{ctx}} = 47, n_{\text{tgt}} = 64$   | 17.1      | 27.7        |           | 0.12095     | 19.8              |      | 0.14352     | 1.013   |
| $n_{\text{ctx}} = 47, n_{\text{tgt}} = 128$  | 18.3      | 30.6        |           | 0.11393     | 21.1              |      | 0.13612     | 1.024   |
| $n_{\text{ctx}} = 128, n_{\text{tgt}} = 128$ | 24.8      | 47.1        |           | 0.07313     | 22.9              |      | 0.17226     | 1.052   |
| $n_{\text{ctx}} = 128, n_{\text{tgt}} = 256$ | 28.6      | 51.2        |           | 0.06499     | 23.0              |      | 0.16480     | 1.007   |

The table gives a clean reading. First, the RBF behavior itself is strong: the model’s MSE is close to KRR in all four evaluations. Second, the rank requirement is not the strict operator rank. Under the prediction-risk definition, the relevant ranks are roughly 17–29, not 28–51. Thus it is not paradoxical that a compact representation could support near-KRR prediction while failing to recover every strict tail mode.

Third, the final residual stream contains substantial RBF-KRR structure. At the prediction-risk rank, the raw final-state basis already reduces the KRR operator error to 0.06–0.12, and the full rank curves continue down to exact operator recovery when the retained basis spans the context-position space. This is an availability result: the information needed to build the RBF KRR operator is present in the final context states to a significant degree. It is weaker than the linear certificate, where the reduced-operator error is essentially numerical zero at the task rank, but it is far from a failure.

The unresolved part is the causal seed extractor. Its dimension is close to  $r_{5\%}$  in the  $n_{\text{ctx}} = 47$  runs and slightly short in the  $n_{\text{ctx}} = 128$  runs, but the induced operator error remains around 0.14–0.17. This exceeds the implied prediction-risk operator tolerance. Therefore the RBF causal

span should not be claimed as a complete KRR mechanism. The right interpretation is narrower and cleaner:

RBF behavior and final-state availability succeed,

while

causal seed extraction remains only partially aligned with the KRR operator.

This distinction is useful. It says the RBF checkpoint has learned near-KRR prediction and exposes much of the relevant operator structure, but our current layerwise causal basis construction is not yet a fully satisfactory mechanistic certificate for the nonlinear task.

## 11 Experiment 4: Causal Direction Removal

The fourth experiment asks whether the task-important directions inside  $Q$  are actually used by the transformer. This is necessary because decodability and sufficiency do not by themselves prove causal use.

Start with a raw final basis  $Q = [q_1, \dots, q_r]$ . For each direction, form  $Q_{-j}$  by removing that column. The point of the leverage score is to rank directions by their importance to the KRR-constrained reduced operator, not by activation magnitude. A direction should count as task-important if deleting it changes  $T_Q$ 's response on task-label probes. Define

$$\Lambda_j^{\text{red}}(U) := \left( \frac{\sum_{u \in U} \|(T_Q - T_{Q_{-j}})u\|_2^2}{\sum_{u \in U} \|Tu\|_2^2 + \eta_{\text{num}}} \right)^{1/2}.$$

A direction has high task leverage if this score is large on task-label probes. This ranking is what connects the causal intervention to the operator certificate: the surgery removes directions that  $T_Q$  itself says are important for task-distributed ridge responses.

The intervention removes selected directions from the context-token residual stream:

$$H_s^{\text{ctx}} \mapsto H_s^{\text{ctx}} - \sum_{j \in J} q_j q_j^\top A H_s^{\text{ctx}},$$

then rolls the remaining layers forward. Matched controls remove low-task-leverage directions from the same basis, high-isotropic-but-low-task directions, high-activation-variance-but-low-task directions, and random subsets.

Before surgery, the checkpoint is in the task-local certificate regime:

$$E_{\text{task}}(F, T) = 0.00926, \quad E_{\text{task}}(F, T_Q) = 0.00920, \quad E_{\text{task}}(T_Q, T) = 0.00086.$$

The same episodes retain the off-distribution drift seen in Experiment 1:

$$E_{\text{iso}}(F, T) = 0.266, \quad E_{\text{iso}}(T_Q, T) = 0.0378.$$

At residual state  $s = 1$ , removing a single high-task-leverage direction causes prediction drift 4.68, KRR-error inflation 649 $\times$ , and task-response damage 41.59. Removing a single low-task-leverage direction from the same basis is nearly inert: prediction drift  $9.8 \cdot 10^{-4}$ , KRR-error inflation 1.03 $\times$ , and task-response damage 0.0024. The layer sweep localizes the effect: high-task surgery is most damaging at early residual states, before later attention has routed the relevant context information into target tokens, and context-only surgery after the final layer is a no-op. At  $s = 1$ , aggregate task leverage and task response damage are strongly related across matched non-random controls (log-log Pearson 0.91, Spearman 0.87).

The complement ablation sharpens the mediation claim. At the same residual state  $s = 1$ , keeping only the  $A$ -projection of context residual states onto  $Q$ ,

$$H_s^{\text{ctx}} \mapsto QQ^\top AH_s^{\text{ctx}},$$

nearly preserves the task-local response over 8 episodes:

$$E_{\text{task}}(F, T) = 0.01866, \quad \text{response damage} = 0.01659.$$

Removing  $Q$  instead,

$$H_s^{\text{ctx}} \mapsto H_s^{\text{ctx}} - QQ^\top AH_s^{\text{ctx}},$$

destroys the response:

$$E_{\text{task}}(F, T) = 145.43, \quad \text{response damage} = 145.43.$$

So the complement of  $Q$  is largely dispensable for the task-local KRR response in this intervention family, while  $Q$  itself is necessary.

Whole-subspace intervention at residual state  $s = 1$

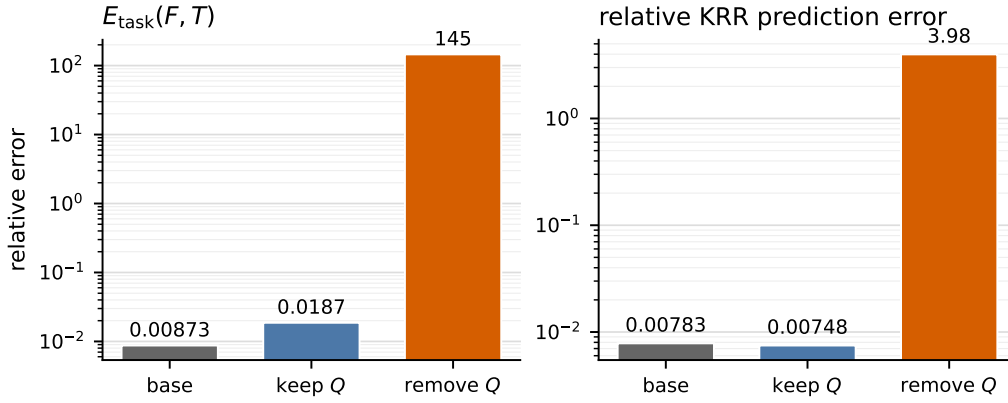


Figure 4: Experiment 4: whole-subspace complement ablation at residual state  $s = 1$ . Left panel: task finite-difference response error  $E_{\text{task}}(F, T)$ . Right panel: relative KRR prediction error  $\|F_{\text{surg}}(y) - Ty\|_2 / \|Ty\|_2$ . Keeping only the  $Q$ -projection nearly preserves both quantities; removing  $Q$  destroys both.

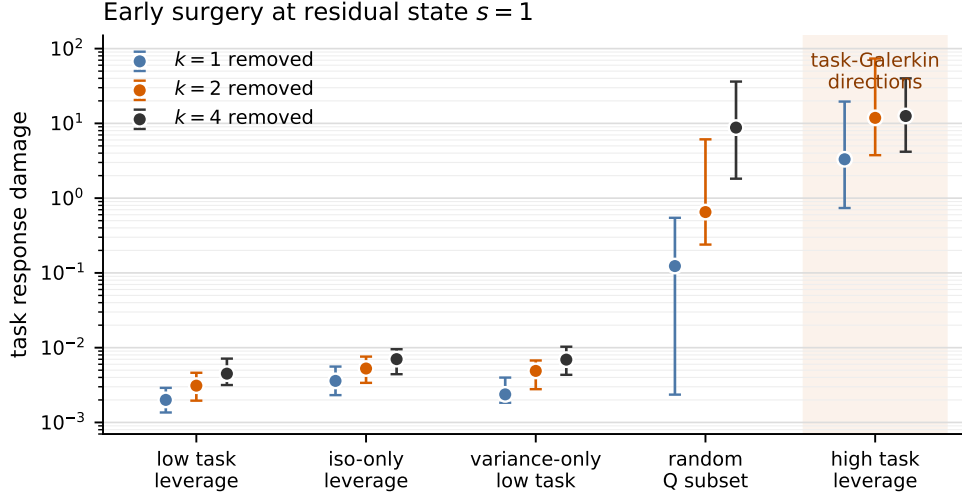


Figure 5: Experiment 4: causal effect of task-important activation directions. The plotted metric is task finite-difference response damage  $(\sum_j \|D_\epsilon F_{\text{surg}}(y)[u_j] - D_\epsilon F_{\text{base}}(y)[u_j]\|^2 / (\sum_j \|Tu_j\|^2 + \eta_{\text{num}}))^{1/2}$ , for task probes  $u_j \sim \mathcal{N}(0, A)$ . High task-leverage directions produce orders-of-magnitude larger damage than low-task-leverage directions from the same activation basis.

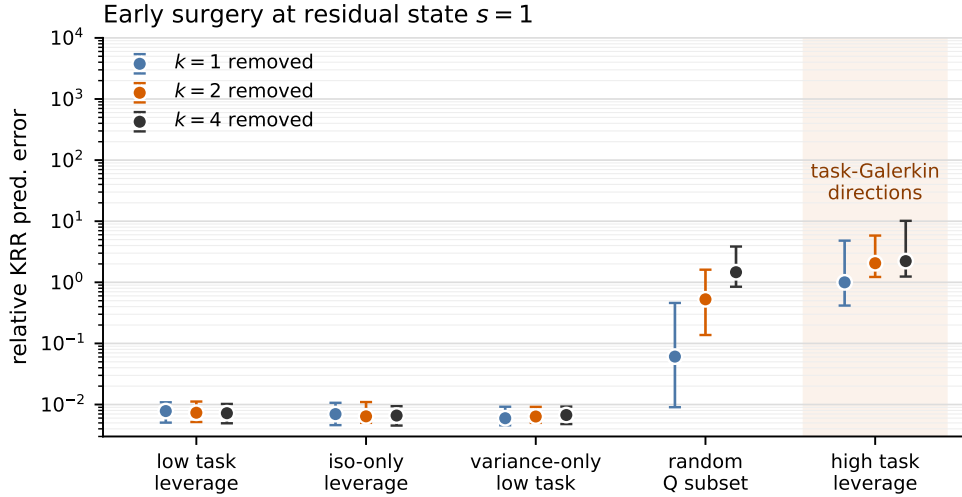


Figure 6: Experiment 4: the same surgery plot, but measuring relative KRR prediction error  $\|F_{\text{surg}}(y) - Ty\|_2 / \|Ty\|_2$  instead of finite-difference response damage. High task-leverage removals also destroy the model’s prediction, while low-task-leverage removals are nearly inert.



This supports the causal-use claim in an operational sense. The directions that the reduced KRR operator identifies as task-important are not merely decodable; removing them selectively damages the transformer’s task-local response, while matched controls do not.

## 12 What Is Justified

The experiments justify a bounded mechanistic claim:

For the tested checkpoint, geometries, perturbation scale, and task-label probe distribution, the transformer’s local label response is well approximated by a reduced KRR operator induced by a compact context-side activation subspace. In successful settings the accepted seed span is on the scale predicted by the task-visible KRR spectrum and its induced reduced operator is close to exact KRR; its high-task-leverage directions are causally important for the transformer’s response.

They do not justify stronger claims. They do not show that the transformer globally equals ridge regression. They do not show that derivative matching at one sampled label vector implies equality on a neighborhood. They do not show that the transformer literally executes the explicit algorithm

$$y \mapsto QQ^\top y \mapsto K_t QQ^\top y$$

as its internal program. They also do not show that architecture alone guarantees the behavior. The task-visible rank calculation supplies a requirement for context-side subspaces; it is not an SGD theorem.

The clean final formulation is:

The trained transformer locally realizes the ridge-regression label-response operator on task-distributed directions. In the tested regime, this behavior is explained by a compact context-side activation subspace whose induced reduced ridge operator approximates exact KRR, whose dimension is on the task-visible scale in successful settings, and whose high-leverage directions are causally important for the transformer’s response.

## 13 Logical Chain

The argument can be read as a sequence of commitments, each narrower than the previous one:

1. Freeze the episode geometry  $G = (X_{\text{ctx}}, X_{\text{tgt}})$ .
2. Under fixed  $G$ , ridge regression becomes  $T = K_t A^{-1}$ .
3. Treat the transformer as a label map  $F : \mathbb{R}^{n_{\text{ctx}}} \rightarrow \mathbb{R}^{n_{\text{tgt}}}$ .
4. Estimate the local response  $D_\varepsilon F(y)[u]$ .
5. Test behavior by checking  $E_\varepsilon(F, T)$  on task-label probes.
6. Compute the task-visible rank from  $C_G = K_t A^{-1/2}$ .
7. Extract activation-derived context-side bases  $Q$ .
8. Construct the KRR-constrained reduced operator  $T_Q = K_t QQ^\top$ .

9. Test subspace sufficiency with  $E(T_Q, T)$ .
10. Test model-subspace alignment with  $E_\varepsilon(F, T_Q)$ .
11. Compare seed layerwise span dimension  $d_{\text{seed}}$  to  $r_{\text{task}}$ .
12. Remove high-leverage  $Q$ -directions and check whether task-local response is damaged.

This is the full mechanism story: behavioral similarity, a KRR-structured activation certificate, rank-scale agreement with the task-visible spectrum, and causal specificity of the high-leverage directions.